



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE(Advanced)
REVIEW TEST
02-01-2022

JEE(Main + Advanced) : NURTURE COURSE [PHASE : S(TEPS)]

Time : 3 Hours

PAPER-1

Maximum Marks : 198

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **28** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140, Cu = 63.5, Ne = 20, K = 39, Mn = 55

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I(i) : (Maximum Marks: 18)

- This section contains **SIX (06)** questions.
- Each question has Four options (A), (B), (C) and (D). ONLY ONE of these four options is correct.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If ONLY the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

-
1. A equilateral triangular loop is carrying steady current I and magnitude of its magnetic dipole moment is M . If this equilateral triangular loop changed to a circular loop and it carries the same current, the magnitude of the magnetic dipole moment of circular loop will be.....

(A) $\frac{M}{\pi}$

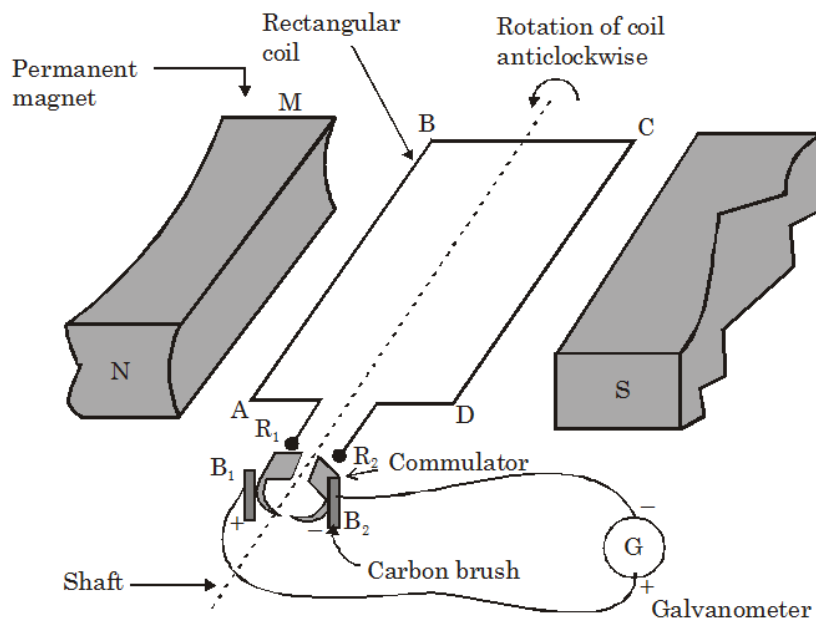
(B) $\frac{9M}{\pi}$

(C) $\frac{3M}{\pi}$

(D) $\frac{3\sqrt{3}M}{\pi}$

Space for Rough Work

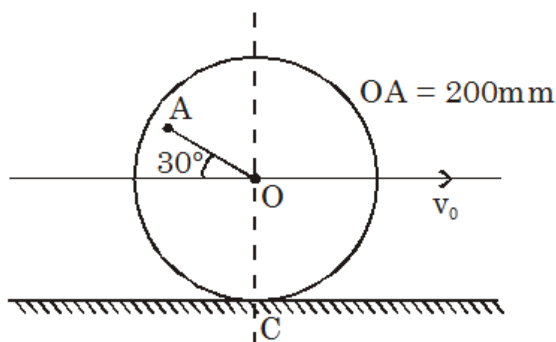
2. A coil is being rotated in a uniform magnetic field to make a generator. The coil is being rotated with a constant angular velocity by a mechanism not shown in the figure. The commutator is connected to the rotating coil and rotates with the coil at the same angular velocity. It is connected to the external load (galvanometer in this figure) by the means of two carbon brushes. The commutator arrangement makes sure that the current in the circuit is unidirectional even if the emf induced across the coil is sinusoidal in nature. The current in galvanometer is considered +ve if it moves from + to – terminal within the galvanometer. Choose the correct statement :



- (A) At the instant shown, emf in wire AB and CD is zero
- (B) At the instant shown, emf in wire AB and CD is such that A is at a higher potential than B and C is at a higher potential than D
- (C) At the instant shown, emf in wire AB and CD is such that A is at a lower potential than B and C is at a lower potential than D
- (D) At the instant shown, emf in wire AB and CD is such that A is at a higher potential than B and C is at a lower potential than D

Space for Rough Work

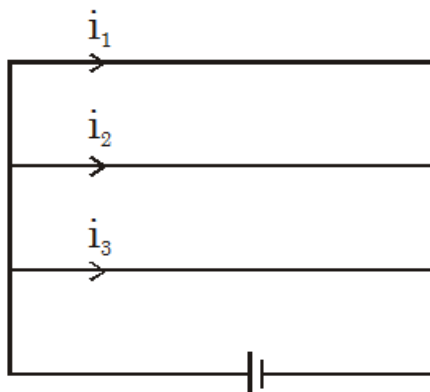
3. The wheel of radius $r = 300$ mm rolls to the right without slipping and has a velocity $v_0 = 3$ m/sec of its centre O. Calculate the velocity of the point A on the wheel for the instant represented in figure.



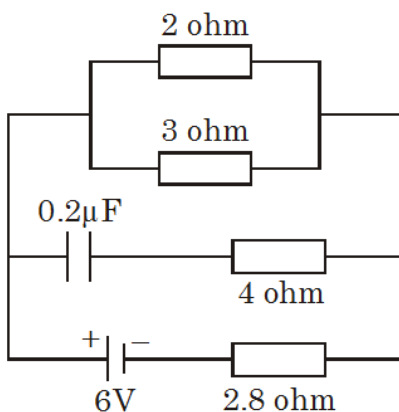
- (A) 4.36 m/sec (B) 5 m/sec (C) 3 m/sec (D) 3.72 m/sec
4. The capacities of two condensers were compared using an electrometer. The capacitors were charged to potentials $V_1 = 300$ V and $V_2 = 100$ V. They were connected in parallel with positive terminals joined together. The p.d between the plates measured by electrometer was 250 V. Find the ratio of capacitance $\frac{C_1}{C_2}$.
- (A) 3 : 1 (B) 2 : 1 (C) 1 : 3 (D) 4 : 1

Space for Rough Work

5. Three long wires of resistance in the ratio 3 : 4 : 5 are connected parallel to each other across a battery of negligible internal resistance. What is the ratio of the distances of middle wire from the others if the net force on it is zero ?



- (A) 5 : 3 (B) 4 : 5 (C) 3 : 4 (D) 1 : 2
6. The steady state current in the 2 ohm resistor shown in figure neglecting the internal resistance of battery will be



- (A) 1 ampere (B) 0.9 ampere (C) 0.8 ampere (D) 1.4 ampere

Space for Rough Work

SECTION-I(ii) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

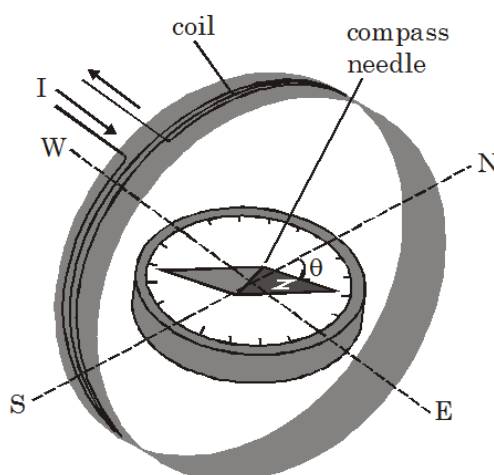
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

7. A tangent galvanometer is an old form of ammeter, consisting of an ordinary magnetic compass mounted at the centre of a coil whose axis is horizontal and oriented along the east-west (Fig.). If there is no current in the coil, the compass needle points north. The compass is in horizontal plane. The horizontal component of earth's magnetic field is 5×10^{-5} T. There are 100 turns in the coil, each having a radius of 3.14 m. If the current in the coil is I , deflection in the compass needle is θ . Choose the correct combination of I and θ .



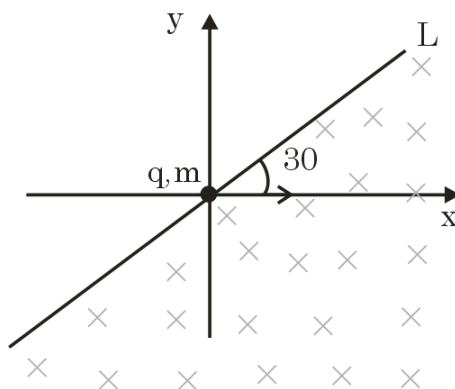
(A) $I = 1\text{ A}$, $\theta = \tan^{-1}(2.5)$ towards east

(B) $I = 2.5\text{ A}$, $\theta = 45^\circ$ towards east

(C) $I = \sqrt{3}\text{ A}$, $\theta = 30^\circ$ towards east

(D) $I = 4\text{ A}$, $\theta = \tan^{-1}\left(\frac{8}{5}\right)$ towards east

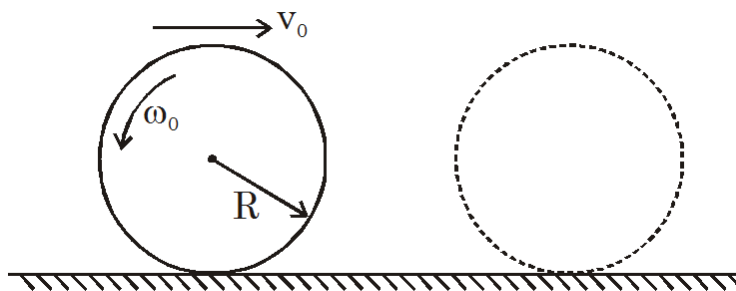
8. A long cylindrical rod has charge per unit volume of ρ . Its radius is R . If it is rotated with an angular velocity ω about central axis along its length, magnetic field at its centre is B_0 . But if it is translated with a velocity of ωR parallel to itself, the magnetic field at its surface is B_1 . In all cases, the observer is at rest.
- (A) $B_0 > B_1$
- (B) $B_0 < B_1$
- (C) When the cylinder is rotated, magnetic field at a point outside the cylinder is zero
- (D) When the cylinder is translated, magnetic field at a point inside it on the axis is zero.
9. A charged particle of charge q and mass m is entering at origin with a velocity $v\hat{i}$ in a uniform magnetic field $-B\hat{k}$ which is extended upto infinite at right of the line L . Select the correct option(s). Given that $\frac{mv}{qB} = 1$ meter, and $v = 1$ m/s.



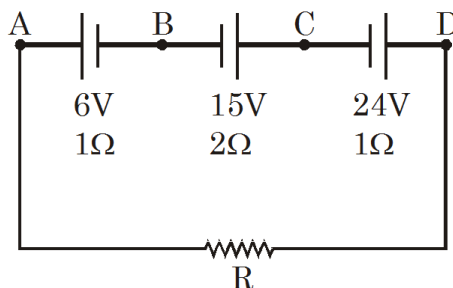
- (A) Velocity of the particle makes an angle 60° with x-axis while it is coming out of the magnetic field.
- (B) The co-ordinates of point of exit is $\left(\frac{1}{2}m, \frac{\sqrt{3}}{2}m\right)$
- (C) The co-ordinates of point of exit is $\left(\frac{\sqrt{3}}{2}m, \frac{1}{2}m\right)$
- (D) Time spent by the particle in the magnetic field is $\frac{\pi}{3}$ sec

Space for Rough Work

10. A solid steel ball is given a linear velocity v_0 along a level surface and at the same time a reverse spin ω_0 . The coefficient of friction between the surfaces is μ .



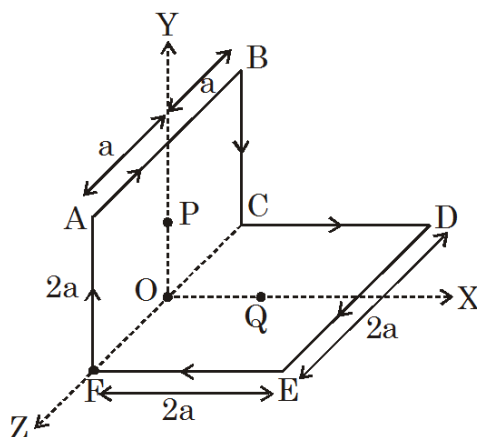
- (A) If $\omega_0 > \frac{5v_0}{R}$, the ball will return to its initial point.
- (B) The ball will stop rotating at $t = \frac{R}{\mu g} \left[\omega_0 - \frac{5v_0}{2R} \right]$
- (C) If $\omega_0 = \frac{2.5v_0}{R}$, its entire mechanical energy will be dissipated in the form of heat
- (D) The velocity of the center will become zero at $t = \frac{v_0}{\mu g}$
11. In the given circuit potential of the point A is 9V higher than potential of the point B. Choose correct alternative(s)



- (A) Value of resistance R is 1Ω
- (B) Value of resistance R is 7Ω
- (C) magnitude of potential difference between B and D is 30 V.
- (D) magnitude of potential difference between B and C is 48 V.

Space for Rough Work

12. A wire frame ABCDEF is kept as shown in the figure. I is the current in the wire frame.



- (A) The direction of magnetic field at 'O' due to the wire frame will be along $\left(\frac{-\hat{i} - \hat{j}}{\sqrt{2}} \right)$
- (B) The direction of magnetic field at P (0, a, 0) due to wire CD and DE will along $-\hat{j}$.
- (C) The direction of magnetic field at Q (a, 0, 0) due to wire EF and CD will be along $(-\hat{j})$.
- (D) The direction of magnetic field at P (0, a, 0) due to wire AF and CB will be along $(-\hat{i})$

Space for Rough Work

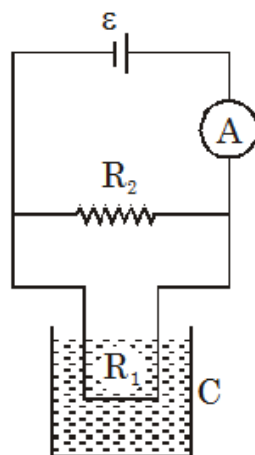
SECTION-II : (Maximum Marks: 24)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

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Zero Marks : 0 In all other cases.

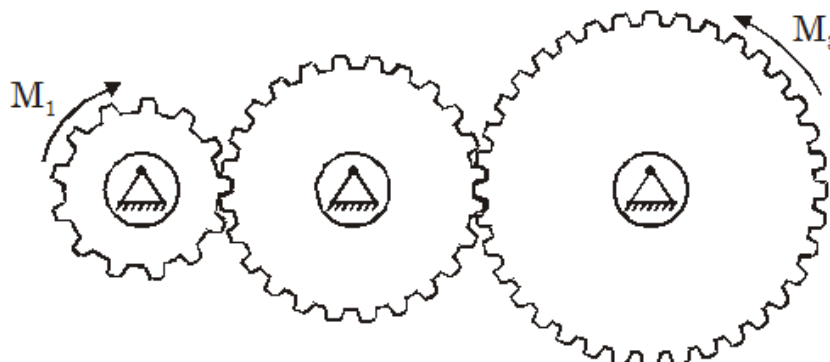
1. The distance between the parallel plates of an isolated charged condenser is $d = 5\text{ cm}$ and the intensity of the field is $E = 300\text{ V cm}^{-1}$. A slab of dielectric constant $K = 5$ and 1 cm wide is inserted parallel to the plates. Determine the potential difference between the plates, after the slab is inserted, (in volt)
2. The coil of a calorimeter C has a resistance of $R_1 = 60\Omega$. The resistance $R_2 = 30\Omega$. The coil R_1 is connected to the circuit as shown in figure. The reading of ammeter is 3 A and 240 gram water is present in container. What will be the change in temperature of water (in $^{\circ}\text{C}$) after 7 minutes . Disregard the resistance of the battery and the ammeter, and the heat losses and heat capacity of the calorimeter and the resistor. (Specific heat of water is $4200\text{ J Kg}^{-1}\text{ K}^{-1}$)



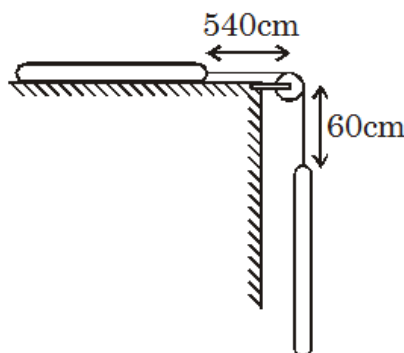
3. A parallel plate capacitor consists of horizontally-situated plate of radius 1 cm , the lower plate is fixed, and the top is plate is hung by a spring of stiffness $k = 36\text{ N/m}$. The distance between the plates is $d = 1 \times 10^{-3}\text{ m}$. When the capacitor is connected by weightless wires to a battery, the spring is extended by $\Delta x = 0.5 \times 10^{-3}\text{ m}$ when it first comes to rest. The plates have very small mass. Determine the emf of the battery. Electric constant $\epsilon_0 = 8.85 \times 10^{-12}\text{ C}^2/\text{Nm}^2$. (Assume that no heat is loss in the circuit)

Space for Rough Work

4. Three gear wheels are connected to each other as shown in figure. The radii of the wheels from left to right are 0.5 m, 2m, and 3m, A torque of 60 N-m is applied to the left most gear wheel. What torque M_3 (in N-m) must be applied to the third wheel so that they do not spin at all ?



5. A man of mass 40 kg is standing on the edge of a horizontal disc of radius 2 m and mass 20 kg. The disc can rotate freely around a vertical axis passing through its centre. With what angular velocity (in rad/s) will the platform rotate, if the man goes along its edge with a speed $V = 2$ m/s relative to the platform ?
6. On a smooth horizontal table there is a metal rod of length $L = 160$ cm. We attached an identical rod to this rod as shown in the diagram. The connection is by a metallic wire whose mass is negligible. The friction is negligible and the system is released at $t = 0$. Find the potential difference between left end of the horizontal rod and the lower end of the vertical rod at $t = 1$ sec. Assume that the rod does not collide with the pulley during the time interval. The horizontal component of earth's magnetic field is 2×10^{-5} T perpendicular to the plane of paper directed away from the reader, and the vertical component of earth's magnetic field is 4×10^{-5} T in the vertically downward direction.



Space for Rough Work

PART-2 : CHEMISTRY
SECTION-I(i) : (Maximum Marks: 18)

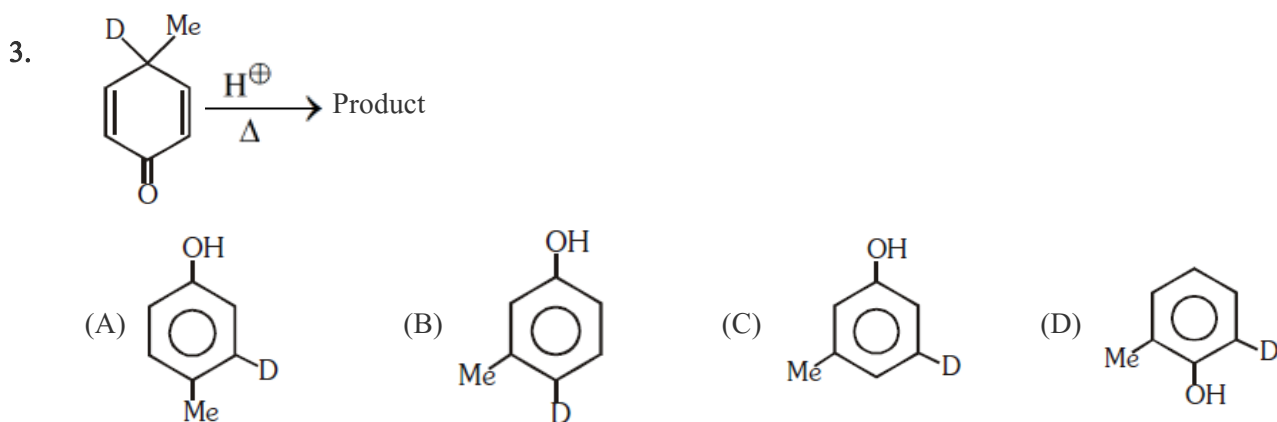
- This section contains **SIX (06)** questions.
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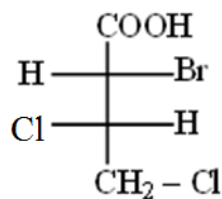
Negative Marks : -1 In all other cases

- Calculate $\Delta S_{\text{universe}}$ for following chemical reaction
 $\text{C (graphite)} + 2\text{H}_2(\text{g}) \rightarrow \text{CH}_4(\text{g}); \Delta_f H^\circ = -74.81 \text{ kJ, at } 298 \text{ K.}$
 The standard entropies of C (graphite), $\text{H}_2(\text{g})$ and $\text{CH}_4(\text{g})$ are 5.740, 130.684 and 186.264 J/K-mol, respectively.
 (A) 170.2 J/K (B) -170.2 J/K (C) 251.0 J/K (D) -80.84 J/K
- $\text{N}_2(\text{g})$ and $\text{H}_2(\text{g})$ are taken in a vessel in mass ratio of 7 : 1. Only reaction $\text{N}_2 + 2\text{H}_2 \rightleftharpoons \text{N}_2\text{H}_4(\text{g})$ occurs. Pressure due to N_2H_4 at equilibrium is 0.2 times of total pressure P. Then at equilibrium
 (A) Partial pressure of $\text{N}_2 = 2P/15$ (B) Partial pressure of $\text{H}_2 = 8P/25$
 (C) $2P_{\text{N}_2} = P_{\text{H}_2}$ (D) $P_{\text{N}_2} = 2P_{\text{H}_2}$



Space for Rough Work

4. Assign R, S configuration for given compounds.



- (A) 2R, 3S (B) 2S, 3S (C) 2S, 3R (D) 2R, 3R
5. The specie among the following that has maximum difference of number of BONDS of central atom between Lewis structure and structure according to VSEPR is.
- (A) H_2SO_4 (B) POF_3 (C) CO_3^{2-} (D) ClO_4^-
6. The pair in which 1st atom has more EA than 2nd atom is.
- (A) O(g), S(g) (B) F(g), Cl(g) (C) P(g), N(g) (D) I(g), Br(g)

Space for Rough Work

SECTION-I(ii) : (Maximum Marks: 24)

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Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

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7. For a reaction, $A \rightleftharpoons P$, equilibrium constant at temperatures T_1 and T_2 are 4 and 2 respectively-

If $T_2 < T_1$, the correct statement(s) is (are) (Assume ΔH° and ΔS° are independent of temperature and ratio of $\ln K$ at T_1 to $\ln K$ at T_2 is greater than T_2/T_1 . Here H, S, G and K are enthalpy, entropy, Gibbs energy and equilibrium constant, respectively.)

(A) $\Delta H^\circ < 0$

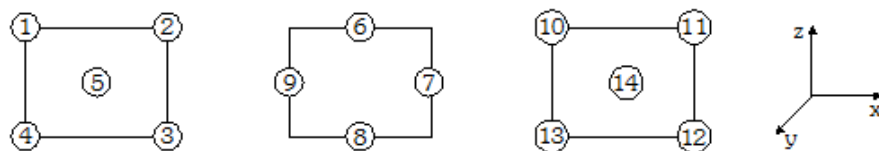
(B) $\Delta G^\circ < 0$ at T_1 K

(C) $\Delta S^\circ > 0$

(D) $\Delta G^\circ > 0$ at T_2 K

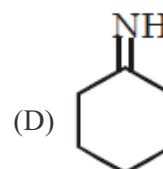
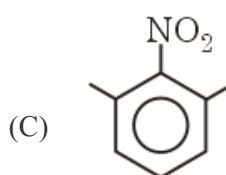
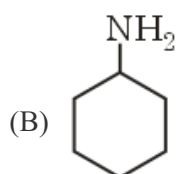
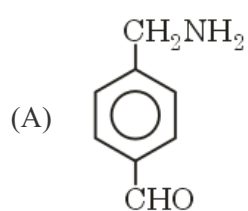
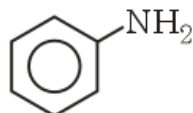
Space for Rough Work

8. A layered sequence for fcc is as shown below



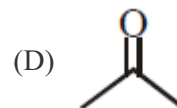
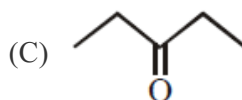
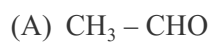
if a face diagonal passes through the centre of atom 4 & the centre(s) of which other atom(s), it passes

- (A) 9, 10 (B) 5, 11 (C) 5, 2 (D) 8, 12
9. Which of the following compound(s) have C-N bond length more than C-N bond length in Aniline ?



Space for Rough Work

10. Select the compound(s) whose enol form show(s) geometrical isomerism?



11. The maximum number of electrons in Ti^{2+} for which $\ell \times m \leq -1$ is.

(A) 4

(B) 5

(C) 6

(D) 7

12. The oxo-acids that will have atleast one oxo-linkage is.



Space for Rough Work

SECTION-II : (Maximum Marks: 24)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
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- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

1. The formation of dextrorotatory isomer A from its levorotatory isomer B in water is reversible reaction having 1st order kinetics in both direction. Racemic mixture is the equilibrium product if reaction starts from either A or B. Analysis of polarimeter experiment showed that 10% of B was converted into A in 230.3 sec at 25°C.

(log 5 = 0.7, log 3 = 0.48 and log 2 = 0.3)

Based on given information select number of correct statement(s) among following.

- (A) Rate constant of forward reaction at 25°C is $5 \times 10^{-4} \text{ s}^{-1}$
- (B) Rate constant of backward reaction at 25°C is $5 \times 10^{-4} \text{ s}^{-1}$
- (C) Activation energy of forward reaction is equal to activation energy of backward reaction.
- (D) Equilibrium constant (K) at 25°C of $B \rightleftharpoons A$ is 1.
- (E) Equilibrium constant increases by increasing temperature.
- (F) Molar entropy of A and B at 25°C must be same.
- (G) $\Delta_r H$ of the process is zero.
- (H) $\Delta_r G^\circ$ of the process is zero.

- (I) $\Delta_r G$ of the process $B \xrightleftharpoons[k_b]{k_f} A$ is zero, initially.

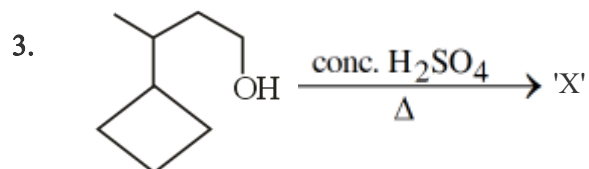
2. An orbital in a H-atom is associated with the following radial and angular parts of its wave function:

$$\Psi(r) = \frac{1}{9\sqrt{30}} \left(\frac{Z}{a_0} \right)^{3/2} \sigma^2 e^{-\sigma/2}; \Psi(\theta, \phi) = \left(\frac{15}{4\pi} \right)^{1/2} \sin\theta \cos\theta \cos\phi$$

Here, $\sigma = \frac{2r}{na_0}$, $a_0 = \frac{h^2}{4\pi^2 m e^2}$, θ = angle from z-axis, ϕ = angle of component in xy plane with x-axis.

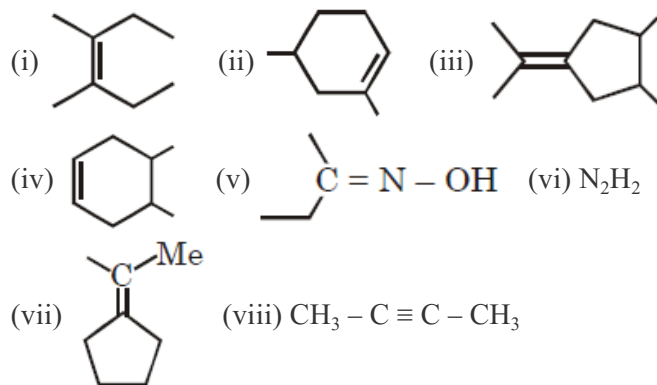
If p_x , p_y & p_z orbitals have been hypothetically assigned m values as +1, -1 & 0 respectively and d_{xy} , d_{yz} , d_{zx} , $d_{x^2-y^2}$ & d_{z^2} orbitals have been hypothetically assigned m values as +2, -2, +1, -1 & 0 respectively, determine value of $(n + \ell + m)$ for the given orbital.

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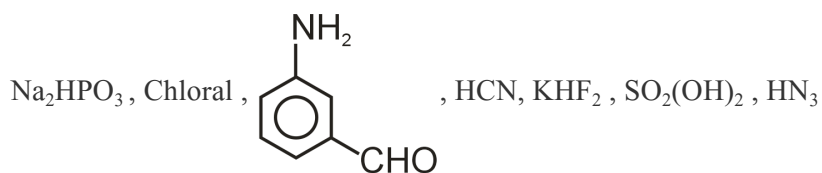
Total number of 1,2 shifts (rearrangements) involve in the above reaction.

4. How many compound(s) can show G.I.?



5. If x = number of π bonds in sodium peroxoborate
 y = number of atoms in 3d series having more than 3 unpaired e^- s
 z = number of 'd' orbitals having two nodal planes
 then $(x + z - y)$ is _____.

6. The number of specie(s) that do not show hydrogen bonding among the following.



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PART-3 : MATHEMATICS
SECTION-I(i) : (Maximum Marks: 18)

- This section contains **SIX (06)** questions.
- Each question has Four options (A), (B), (C) and (D). ONLY ONE of these four options is correct.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If ONLY the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

1. The ordered pair (a, b) such that $x^2 - x - 1$ divides $ax^9 + bx^8 + 1$ is
 (A) (-21, 34) (B) (21, -34) (C) (47, -58) (D) (58, -47)
2. If a, b, c are positive real numbers then minimum value of $\frac{(a+3c)}{(a+c+2b)} + \frac{4b}{(a+b+2c)} - \frac{8c}{(a+b+3c)}$ is
 (A) $7\sqrt{12} + 5$ (B) 3 (C) $8\sqrt{3} - 19$ (D) $12\sqrt{2} - 17$
3. Let $f(n, r)$ be arithmetic mean of minimums of all the 'r' element subset of set $\{1, 2, 3, \dots, n\}$, then $f(100, 20)$ equals
 (A) $\frac{100}{21}$ (B) $\frac{101}{22}$ (C) $\frac{99}{22}$ (D) $\frac{101}{21}$

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4. Following quantities of the given flavours are available in a shop (Ice-creams of the same flavour are identical). 4 Vanilla, 5 Butterscotch, 6 Strawberry, 3 Chocolate, 2 Kaju-pista, then the number of ways in which at least 4 ice-creams can be selected, such that selection of Vanilla is compulsory and Strawberry are taken in pairs or not at all is equal to N then value of remainder when N is divided by 10 is equal to

(A) 2 (B) 4 (C) 6 (D) 8

5. Find polynomials $f(x)$, $g(x)$ and $h(x)$ If they exist such that for all x ,

$$|f(x)| - |g(x)| + h(x) = \begin{cases} -1 & \text{if } x < -1 \\ 3x + 2 & \text{if } -1 \leq x \leq 0 \\ -2x + 2 & \text{if } x > 0 \end{cases}, \text{ the value of } f(3) + g(4) + g\left(\frac{5}{2}\right) \text{ equals}$$

(A) 12 (B) 13 (C) 14 (D) 15

6. If $x = \alpha$ satisfies the equation $\frac{\sin^{-1}(x^{2018}) + 2\cos^{-1}(x^{2017})}{\cos^{-1}(x^{2018}) + 2\sin^{-1}(x^{2017})} = \frac{-5}{2}$

then the value of $\frac{\sin^{-1}(\sin \alpha) + \tan^{-1}(\tan \alpha)}{\cos^{-1}(\cos \alpha) + 2 \operatorname{cosec}^{-1}(\operatorname{cosec} \alpha)}$ is

(A) 2 (B) 1 (C) 0 (D) -1

Space for Rough Work

SECTION-I(ii) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

-
7. 5 alike apples, 4 different mangoes and 3 alike oranges are to be distributed among Honey, Boney and Toney. Number of ways of distribution in which -
- (A) Honey gets exactly 2 apples, one mango and one orange is 384.
- (B) Honey and Boney each gets atleast one apple and atleast one orange is 90
- (C) Boney gets all mangoes is 210
- (D) Toney gets fruits of every species is 2430.
8. Point P lies on $y^2 = kx$ ($k > 0$) such that it divides OA internally in $a : b$ where O is origin and $A(\alpha, \beta)$ is variable point on $x^2 = 4y$ where $\beta < 16$, then choose the correct option(s) -
- (A) sum of all integral values of k is 91
- (B) sum of all integral values of k is 496
- (C) if $k = 4$, then $4 < \beta < 16$
- (D) if $b = 9$ and $k = 2$, then $b - a = 19$. (a & b are coprime numbers)
-

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9. Let set S consists of all the points (x,y) satisfying $9x^2 + 16y^2 \leq 144$. For points in S let maximum and minimum value of $\frac{y-4}{x-9}$ be M and m respectively, then

(A) $M = 1$ (B) $M = \frac{65}{7}$ (C) $m = 1$ (D) $m = \frac{7}{65}$

10. $f(x) = \begin{cases} \log_2(x^2 + 4) & 0 \leq x < 2 \\ \sqrt{9-x} + \sqrt{x-1} & 2 \leq x \leq 4 \end{cases}$, then

(A) $\text{fof}(x) = \sqrt{9 - \log_2(x^2 + 4)} + \sqrt{\log_2(x^2 + 4) - 1}$, if $x \in [0, 2)$

(B) $\text{fof}(x) = \log_2((\sqrt{9-x} + \sqrt{x-1})^2 + 4)$, if $x \in [2, 4]$

(C) $\text{fof}(x) = \sqrt{9 - (\sqrt{9-x} + \sqrt{x-1})} + \sqrt{(\sqrt{9-x} + \sqrt{x-1}) - 1}$, if $x \in [2, 4]$

(D) $\text{fof}(x) = \log_2(\log_2(x^2 + 4) + 4)$, if $x \in [0, 2)$

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11. Let $f : A \rightarrow \mathbb{R}$, $f(x) = \sqrt{(2 - 2^{|x-1|}) \log_x (x^2 - 1)}$, then
(where set A denotes domain and $\mathbb{R}$ denotes set of all real numbers)
- (A) number of integers in domain of $f(x)$ is 1 (B) $f(x)$ is many-one function
(C) domain of function is $(1, 2]$ (D) $f(x)$ is into function
12. Which of the following statements are correct?
where $[.] =$ greatest integer function.
- (A) Range of $\left[x + \frac{1}{2}\right] + \left[x - \frac{1}{2}\right] + 2[-x]$ contains exactly three integers.
(B) Range of $[\sin x] + [\cos x]$ contains exactly two integers.
(C) Range of $\left[x + \frac{1}{2}\right] + \left[x - \frac{1}{2}\right] + 2[-x]$ contains exactly two integers.
(D) Range of $\left[\frac{1}{[x-3]}\right]$ contains exactly three integers

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SECTION-II : (Maximum Marks: 24)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

1. If $a_n = \left(\frac{2n+1}{n+1}\right)^8 C_n$; $n = \{0, 1, 2, \dots, 7, 8\}$
 $b_n = \left(\frac{3n+1}{n+1}\right)^7 C_n$; $n = \{0, 1, 2, \dots, 7\}$
 If $a_p = \max_{0 \leq n \leq 8} (a_n)$ and $b_q = \min_{0 \leq n \leq 7} (b_n)$, where $n \in W$
 Find $(p + q)$
2. Let k denotes the number of integers formed using the digits 0 to 9 (digits used at most once), greater than 999 and less than 10^6 which are divisible by 125, then number of proper factors of k are
3. Consider a family of lines $(4b + 3)x - (b + 1)y = 2b + 1$, where $b \in R$. The minimum area of the triangle which a member of this family with negative gradient makes with the positive semi-axes is equal to

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4. Let a point $P(x, y)$ moving in x - y plane such that it always satisfies $x^2 + y^2 - 8|x| - 8|y| + 31 = 0$ and $\left((1+x^2)^y - 1\right)\left((1+y^2)^x - 1\right) < 0$. If a line passing through $L(-5, 6)$ intersect the locus of P at A, B, C & D in order and $LA \cdot LB \cdot LC \cdot LD = \lambda$ then $\lambda/120$ equals
5. Tangents are drawn from any point on straight line $y = x + 8$ to the auxiliary circle of $x^2 + 16y^2 = 16$. If A and B are the points of contact of these tangent and P, Q are corresponding points of A and B on the ellipse respectively. If locus of mid point of PQ is $x^2 + \alpha y^2 + \beta x + \gamma y = 0$ then value of $(\alpha - \beta + \gamma)$ is
6. Let $f(x)$ is a polynomial in x defined as $f(x) = (1 + x + x^2)(1 + x)^5(1 + x^2)^5$, then coefficient of x^{15} in $f(x)$ is
-

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Space for Rough Work

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions
and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form
number of the candidate, and that question
paper and ORS codes are the same.

Signature of the Invigilator

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